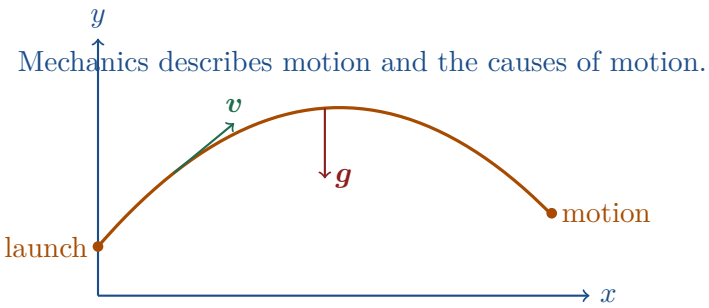
Classical Mechanics for High School Students

Lecture Notes from the Ground Up

Prepared for a high-school mechanics lecture



How to use these notes

These notes are written for high-school students. They do not assume calculus. The goal is to make mechanics learnable by connecting every formula to a clear idea, a diagram, and a worked example. Whenever a formula appears, read it as a sentence about motion, forces, energy, or momentum.

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1 What mechanics is about

Mechanics is the part of physics that studies motion and the causes of motion. It answers questions such as:

- Where is an object after some time?
- How fast is it moving?
- Why does it speed up, slow down, or change direction?
- How can we predict collisions, falling motion, circular motion, and orbits?

In high-school physics, mechanics is usually divided into several parts:

1. **Kinematics**: description of motion, without asking why it happens.
2. **Dynamics**: explanation of motion using forces and Newton's laws.
3. **Energy**: solving motion problems using work, kinetic energy, and potential energy.
4. **Momentum**: solving collision and explosion problems.
5. **Circular motion and gravitation**: motion around a circle and motion caused by gravity.
6. **Oscillations and fluids**: springs, pendulums, pressure, buoyancy.

Model

A **model** is a simplified description of reality. In mechanics we often ignore air resistance, treat objects as points, or assume friction is constant. A model is not reality; it is a useful approximation.

1.1 The most important habit: draw the situation

A large part of mechanics becomes easier if you draw:

- the object,
- the coordinate axes,
- velocity arrows,
- force arrows,
- known and unknown quantities.

Universal problem-solving routine

For most mechanics problems, use this order:

1. Read the question and identify what is asked.
2. Draw a diagram.
3. Write known quantities with units.
4. Choose a coordinate system and signs.
5. Choose the principle: kinematics, Newton's laws, energy, or momentum.
6. Substitute numbers only after writing the symbolic formula.
7. Check units and reasonableness of the answer.

1.2 Physical quantities and units

A physical quantity is a number together with a unit. Writing only the number is usually meaningless.

Quantity	Symbol	SI unit
Length / position	x, s, h	metre, m
Time	t	second, s
Mass	m	kilogram, kg
Velocity	v	metre per second, m/s
Acceleration	a	metre per second squared, $\mathrm{m/s^2}$
Force	F	newton, $\mathrm{N} = \mathrm{kg\,m/s^2}$
Energy / work	E, W	joule, $\mathrm{J} = \mathrm{N\,m}$
Power	P	watt, $\mathrm{W} = \mathrm{J/s}$
Momentum	p	$\mathrm{kg\,m/s}$

Common mistake: forgetting units

The answer 5 is not a physics answer. The answer 5 m/s, 5 N, or 5 J means something. Units are part of the result.

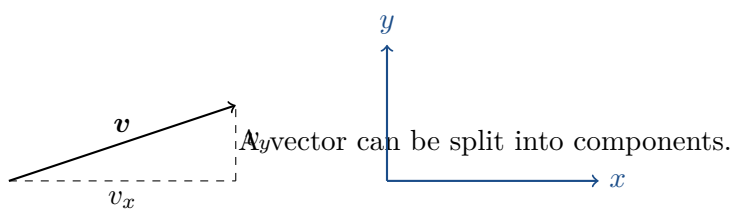
1.3 Scalars and vectors

Some quantities have only size. These are **scalars**. Examples: mass, time, temperature, energy.

Some quantities have size and direction. These are **vectors**. Examples: displacement, velocity, acceleration, force, momentum.

Vector

A vector is a quantity with magnitude and direction. In diagrams it is drawn as an arrow. The length of the arrow represents magnitude; the arrowhead represents direction.



A vector can be split into components.

If a vector has magnitude v and makes an angle θ with the positive x-axis, then

$$v_x = v\cos\theta, \qquad v_y = v\sin\theta.$$

Example 1.1: Components of a velocity

A ball is thrown with speed $v = 20\,\mathrm{m/s}$ at angle 30° above the horizontal. Find the components.

Solution.

$$v_x = v\cos 30^\circ = 20 \cdot 0.866 \approx 17.3\,\mathrm{m/s},$$

$$v_y = v\sin 30^\circ = 20 \cdot 0.5 = 10\,\mathrm{m/s}.$$

The horizontal component is about $17.3\,\mathrm{m/s}$ and the vertical component is $10\,\mathrm{m/s}$.

2 Kinematics in one dimension

Kinematics describes motion without discussing forces. In one dimension, motion happens along one line, for example along a road or vertical line.

2.1 Position, displacement, and distance

Position

The **position** x tells where an object is relative to an origin. It can be positive, negative, or zero.

Displacement

The **displacement** is the change in position:

$$\Delta x = x_{\text{final}} - x_{\text{initial}}.$$

It depends only on the starting and ending points.

Distance

Distance is the total length of the path traveled. Distance is never negative. Displacement can be positive, negative, or zero.

Example 2.1: Distance versus displacement

A student walks from $x = 0\,\text{m}$ to $x = 8\,\text{m}$, then back to $x = 3\,\text{m}$.
Distance traveled:

$$8\,\text{m} + 5\,\text{m} = 13\,\text{m}.$$

Displacement:

$$\Delta x = 3\,\text{m} - 0\,\text{m} = 3\,\text{m}.$$

The student walked $13\,\text{m}$, but ended only $3\,\text{m}$ from the start.

2.2 Average velocity and average speed

Average velocity

Average velocity is displacement divided by time:

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t}.$$

It has a sign, because displacement has a sign.

Average speed

Average speed is distance divided by time:

$$\text{speed}_{\text{avg}} = \frac{\text{distance}}{\Delta t}.$$

It is never negative.

Example 2.2: Average velocity

A cyclist moves from $x = 10\,\mathrm{m}$ to $x = 70\,\mathrm{m}$ in $12\,\mathrm{s}$.

$$\Delta x = 70 - 10 = 60\,\mathrm{m},$$

$$v_{\mathrm{avg}} = \frac{60\,\mathrm{m}}{12\,\mathrm{s}} = 5\,\mathrm{m/s}.$$

2.3 Acceleration

Average acceleration

Acceleration tells how quickly velocity changes:

$$a_{\mathrm{avg}} = \frac{\Delta v}{\Delta t} = \frac{v_{\mathrm{final}} - v_{\mathrm{initial}}}{\Delta t}.$$

If acceleration is positive, velocity changes in the positive direction. If acceleration is negative, velocity changes in the negative direction. Negative acceleration does not always mean slowing down; it depends on the direction of velocity.

Common mistake: negative acceleration

If an object moves left with velocity $v < 0$ and has acceleration $a < 0$, it is speeding up to the left. Negative acceleration does not automatically mean braking.

2.4 Motion with constant acceleration

When acceleration is constant, we can use four important formulas:

Constant acceleration equations

$$v = v_0 + at$$

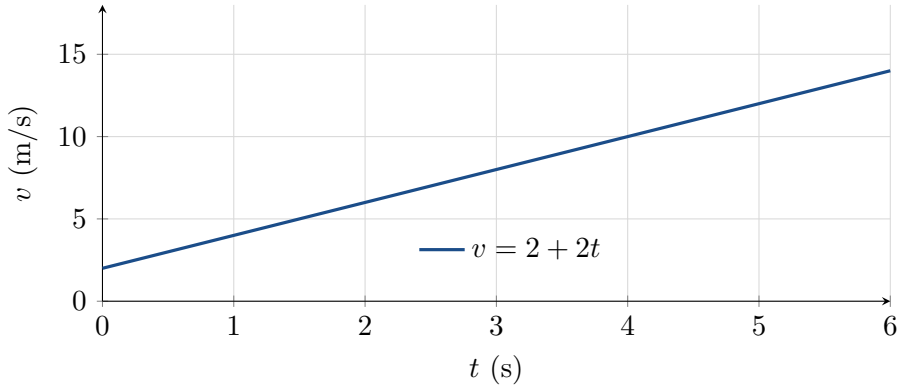
$$x = x_0 + v_0 t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$x - x_0 = \frac{v_0 + v}{2}t$$

Here x_0 and v_0 are initial position and initial velocity.

These formulas are not magic. They come from the idea that constant acceleration changes velocity at a steady rate.



The slope of the velocity-time graph is acceleration. The area under the velocity-time graph is displacement.

Example 2.3: Car accelerating from rest

A car starts from rest and accelerates at $2\,\mathrm{m/s^2}$ for $8\,\mathrm{s}$. Find its final speed and distance traveled.
Known:

$$v_0 = 0, \qquad a = 2\,\mathrm{m/s^2}, \qquad t = 8\,\mathrm{s}.$$

Final speed:

$$v = v_0 + at = 0 + 2 \cdot 8 = 16\,\mathrm{m/s}.$$

Distance:

$$x - x_0 = v_0 t + \frac{1}{2}at^2 = 0 + \frac{1}{2} \cdot 2 \cdot 8^2 = 64\,\mathrm{m}.$$

Example 2.4: Braking distance

A car moves at $20\,\mathrm{m/s}$ and brakes with acceleration $a = -5\,\mathrm{m/s^2}$. How far does it travel before stopping?
Use the formula without time:

$$v^2 = v_0^2 + 2a\Delta x.$$

At stopping, $v = 0$:

$$0 = 20^2 + 2(-5)\Delta x.$$

$$0 = 400 - 10\Delta x, \qquad \Delta x = 40\,\mathrm{m}.$$

2.5 Free fall

Near Earth's surface, if we ignore air resistance, all objects fall with approximately the same acceleration:

$$g \approx 9.81\,\mathrm{m/s^2}.$$

Often in school problems we use $g \approx 10\,\mathrm{m/s^2}$ for easier calculation.

If we choose upward as positive, then acceleration in free fall is

$$a = -g.$$

If we choose downward as positive, then

$$a = +g.$$

Both choices are valid. The important thing is to be consistent.

Example 2.5: Dropping a stone

A stone is dropped from rest from a height of 20 m. Ignore air resistance and use $g = 10\,\mathrm{m/s^2}$. How long does it take to fall?
Choose downward as positive. Then $v_0 = 0$, $a = g = 10\,\mathrm{m/s^2}$, and $s = 20\,\mathrm{m}$.

$$s = v_0 t + \frac{1}{2} g t^2.$$

$$20 = 0 + \frac{1}{2} \cdot 10 \cdot t^2 = 5t^2.$$

$$t^2 = 4, \qquad t = 2\,\mathrm{s}.$$

2.6 Exercises

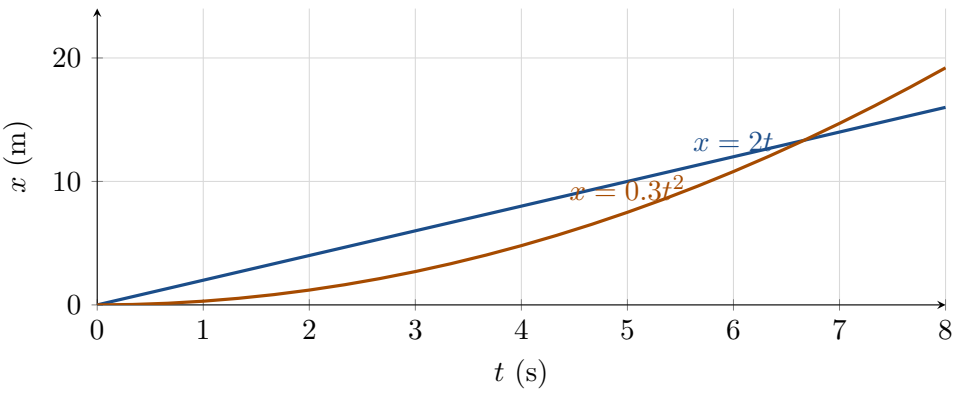
1. A runner moves from $x = 5\,\mathrm{m}$ to $x = 45\,\mathrm{m}$ in $8\,\mathrm{s}$. Find average velocity.
2. A bicycle accelerates from $3\,\mathrm{m/s}$ to $11\,\mathrm{m/s}$ in $4\,\mathrm{s}$. Find acceleration.
3. A car starts from rest and accelerates at $1.5\,\mathrm{m/s^2}$ for $10\,\mathrm{s}$. Find final speed and distance.
4. A ball is thrown upward with speed $20\,\mathrm{m/s}$. Using $g = 10\,\mathrm{m/s^2}$, find the time to reach the highest point.

3 Graphs of motion

Graphs are not decoration. They are a language for motion.

3.1 Position-time graphs

A position-time graph tells where an object is at each time. The slope of a position-time graph is velocity.

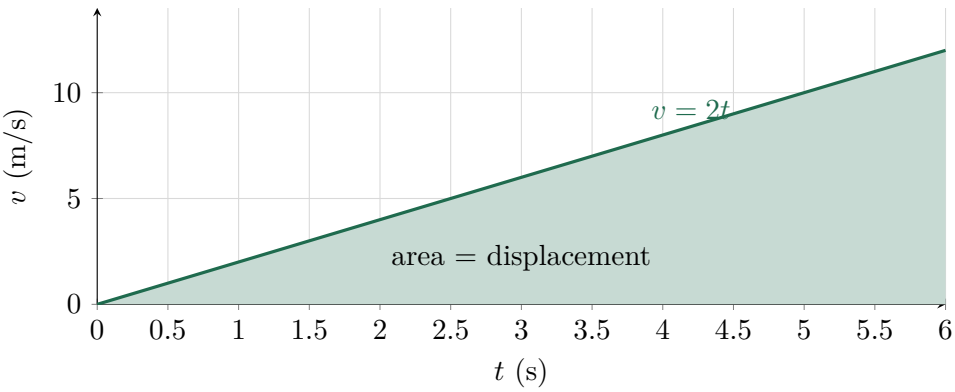


A straight line on an $x(t)$ graph means constant velocity. A curved line means changing velocity.

3.2 Velocity-time graphs

On a velocity-time graph:

- slope means acceleration,
- area under the graph means displacement.



For the graph $v = 2t$ from $t = 0$ to $t = 6\,\mathrm{s}$, the displacement is the area of a triangle:

$$\Delta x = \frac{1}{2} \cdot 6 \cdot 12 = 36\,\mathrm{m}.$$

3.3 Acceleration-time graphs

On an acceleration-time graph, the area under the graph is the change in velocity:

$$\Delta v = a\Delta t$$

when acceleration is constant.

Example 3.1: Reading a velocity graph

A velocity-time graph is a straight line from $v = 4\,\mathrm{m/s}$ at $t = 0$ to $v = 12\,\mathrm{m/s}$ at $t = 4\,\mathrm{s}$. Acceleration:

$$a = \frac{12 - 4}{4} = 2\,\mathrm{m/s^2}.$$

Displacement equals area under the graph, a trapezoid:

$$\Delta x = \frac{4 + 12}{2} \cdot 4 = 32\,\mathrm{m}.$$

Common mistake: confusing graph height and slope

On a position-time graph, the height is position and the slope is velocity. On a velocity-time graph, the height is velocity and the slope is acceleration.

4 Motion in two dimensions

In two dimensions, we describe motion using horizontal and vertical components. The key idea is simple:

Independence of perpendicular directions

Horizontal motion and vertical motion can be studied separately, then combined.

4.1 Position and velocity as vectors

Position in a plane can be written as

$$\boldsymbol{r} = (x, y).$$

Velocity can be written as

$$\boldsymbol{v} = (v_x, v_y).$$

Acceleration can be written as

$$\boldsymbol{a} = (a_x, a_y).$$

4.2 Projectile motion

Projectile motion is the motion of an object thrown into the air, when only gravity acts on it. We ignore air resistance.

If the object starts at the origin with speed v_0 and angle θ, then

$$v_{0x} = v_0 \cos\theta, \qquad v_{0y} = v_0 \sin\theta.$$

The horizontal acceleration is zero:

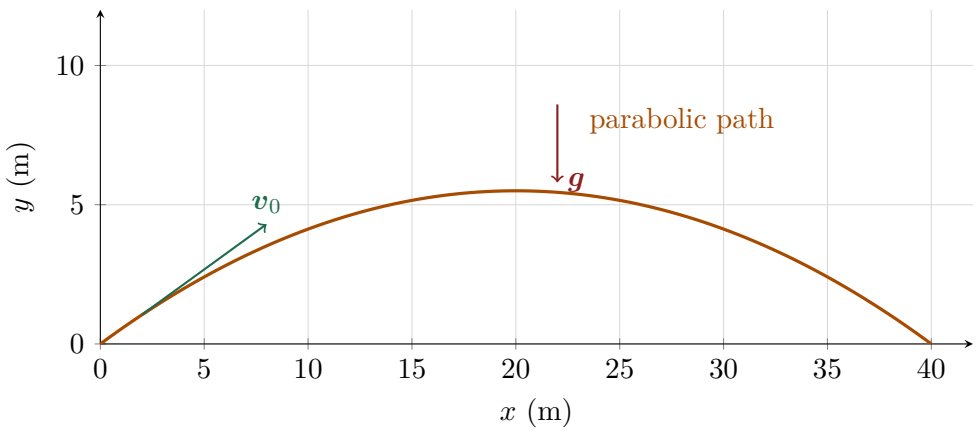
$$a_x = 0.$$

The vertical acceleration is gravity:

$$a_y = -g.$$

Therefore:

$$x(t) = v_0 \cos\theta\, t,$$
$$y(t) = v_0 \sin\theta\, t - \frac{1}{2} g t^2.$$



4.3 Time of flight, maximum height, and range

For launch and landing at the same height:

Time of flight:

$$T = \frac{2v_0 \sin\theta}{g}.$$

Maximum height:

$$H = \frac{v_0^2 \sin^2\theta}{2g}.$$

Range:

$$R = \frac{v_0^2 \sin(2\theta)}{g}.$$

These formulas assume that the projectile lands at the same vertical level from which it was launched.

Example 4.1: Projectile launched at $30°$

A ball is thrown with speed $20\,\mathrm{m/s}$ at $30°$. Use $g = 10\,\mathrm{m/s^2}$.
Initial components:

$$v_{0x} = 20\cos 30° \approx 17.3\,\mathrm{m/s},$$

$$v_{0y} = 20\sin 30° = 10\,\mathrm{m/s}.$$

Time of flight:

$$T = \frac{2v_{0y}}{g} = \frac{2\cdot 10}{10} = 2\,\mathrm{s}.$$

Range:

$$R = v_{0x}T \approx 17.3\cdot 2 = 34.6\,\mathrm{m}.$$

Maximum height:

$$H = \frac{v_{0y}^2}{2g} = \frac{10^2}{20} = 5\,\mathrm{m}.$$

Common mistake: using the same equation in both directions

In projectile motion, horizontal and vertical directions have different accelerations. Horizontally, $a_x = 0$. Vertically, $a_y = -g$.

5 Forces and Newton's laws

Kinematics describes motion. Dynamics explains motion using forces.

Force

A force is an interaction that can change the motion of an object. Force is a vector. Its SI unit is the newton:

$$1\,\mathrm{N} = 1\,\mathrm{kg.m/s^2}.$$

5.1 Newton's first law

Newton's first law

If the net force on an object is zero, the object remains at rest or moves with constant velocity.

The first law says that motion does not require a force to continue. A force is required to change velocity.

5.2 Newton's second law

Newton's second law

The net force equals mass times acceleration:

$$\sum \boldsymbol{F} = m\boldsymbol{a}.$$

In one dimension:

$$\sum F = ma.$$

5.3 Newton's third law

Newton's third law

If object A exerts a force on object B, then object B exerts a force of equal magnitude and opposite direction on object A.

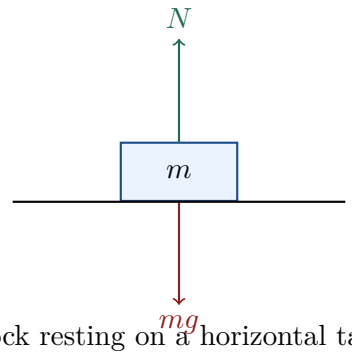
The two forces act on different objects, so they do not cancel each other in the same free-body diagram.

5.4 Common forces

Force	Symbol	Typical formula
Weight	F_g or mg	$F_g = mg$
Normal force	N	depends on situation
Friction	f	$f_k = \mu_k N$ for kinetic friction
Tension	T	depends on situation
Spring force	F_s	$F_s = kx$ in magnitude
Air resistance	F_d	model-dependent

5.5 Free-body diagrams

A free-body diagram shows all forces acting on one chosen object.



Block resting on a horizontal table

For the block on the table, if it is not accelerating vertically:

$$N - mg = 0,$$

so

$$N = mg.$$

Common mistake: assuming $N = mg$ always

The normal force equals mg only in simple horizontal situations with no vertical acceleration and no other vertical forces. On an incline or in an elevator, N may be different.

5.6 Friction

Friction opposes relative motion or the tendency of relative motion between surfaces.

Kinetic friction:

$$f_k = \mu_k N.$$

Static friction adjusts up to a maximum value:

$$f_s \leq \mu_s N.$$

Example 5.1: Pulling a box with friction

A 10 kg box is pulled horizontally with force 40 N. The coefficient of kinetic friction is $\mu_k = 0.2$. Use $g = 10\,\mathrm{m/s^2}$. Find acceleration.

Weight:

$$mg = 10 \cdot 10 = 100\,\mathrm{N}.$$

Normal force on a horizontal floor:

$$N = 100\,\mathrm{N}.$$

Friction:

$$f_k = \mu_k N = 0.2 \cdot 100 = 20\,\mathrm{N}.$$

Net horizontal force:

$$F_{\text{net}} = 40 - 20 = 20\,\mathrm{N}.$$

Acceleration:

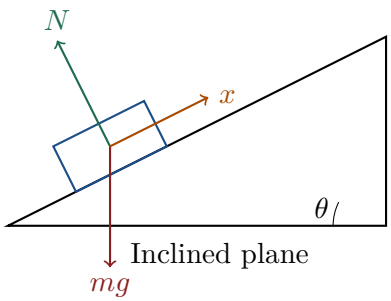
$$a = \frac{F_{\text{net}}}{m} = \frac{20}{10} = 2\,\mathrm{m/s^2}.$$

5.7 Inclined plane

For a block on an incline of angle θ, the weight has two useful components:

$$mg\sin\theta \quad \text{down the slope,}$$

$$mg\cos\theta \quad \text{perpendicular to the slope.}$$



If there is no friction, acceleration down the slope is

$$a = g\sin\theta.$$

Example 5.2: Sliding down a frictionless incline

A block slides down a frictionless incline with $\theta = 30°$. Use $g = 10\,\mathrm{m/s^2}$.

$$a = g\sin 30° = 10 \cdot 0.5 = 5\,\mathrm{m/s^2}.$$

6 Equilibrium and torque

An object is in translational equilibrium if it has no acceleration of its center of mass.

Translational equilibrium

$$\sum \boldsymbol{F} = 0.$$

This means forces balance.

But an extended object can also rotate. To describe rotational tendency, we use torque.

Torque

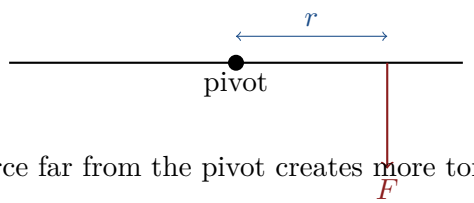
Torque measures the turning effect of a force:

$$\tau = rF \sin\theta.$$

Here r is the distance from the pivot to the point where the force is applied, and θ is the angle between $\boldsymbol{r}$ and $\boldsymbol{F}$.

If the force is perpendicular to the lever arm:

$$\tau = rF.$$



A force far from the pivot creates more torque.

Rotational equilibrium

$$\sum \tau = 0.$$

This means clockwise torques balance counterclockwise torques.

Example 6.1: Balancing a seesaw

A 30 kg child sits 2 m from the pivot. Where should a 60 kg child sit on the other side to balance?
Use torques. Since both weights include the same factor g, it cancels:

$$30g \cdot 2 = 60g \cdot r.$$

$$60 = 60r, \qquad r = 1\,\mathrm{m}.$$

The heavier child must sit closer to the pivot.

6.1 Center of mass

The center of mass is the average position of mass. For two point masses on a line:

$$x_{\mathrm{cm}} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}.$$

Example 6.2: Center of mass

Mass 2 kg is at $x = 0\,\mathrm{m}$, and mass 6 kg is at $x = 4\,\mathrm{m}$. Find the center of mass.

$$x_{\mathrm{cm}} = \frac{2 \cdot 0 + 6 \cdot 4}{2 + 6} = \frac{24}{8} = 3\,\mathrm{m}.$$

The center of mass is closer to the heavier mass.

7 Work, energy, and power

Energy methods often make mechanics problems shorter. Instead of tracking every force over time, we compare energy at the beginning and at the end.

7.1 Work

Work

If a constant force F acts along the direction of displacement s, then

$$W = Fs.$$

If the force makes an angle θ with displacement:

$$W = Fs\cos\theta.$$

Work is measured in joules:

$$1\,\mathrm{J} = 1\,\mathrm{N.m}.$$

Positive work increases kinetic energy. Negative work decreases kinetic energy.

Example 7.1: Work done by a pulling force

A person pulls a crate with force 50 N over distance 6 m in the same direction as motion.

$$W = Fs = 50 \cdot 6 = 300\,\mathrm{J}.$$

7.2 Kinetic energy

Kinetic energy

Kinetic energy is energy of motion:

$$E_k = \frac{1}{2}mv^2.$$

If speed doubles, kinetic energy becomes four times larger, because of v^2.

Example 7.2: Kinetic energy

A 2 kg object moves at 5 m/s.

$$E_k = \frac{1}{2} \cdot 2 \cdot 5^2 = 25\,\mathrm{J}.$$

7.3 Work-energy theorem

Work-energy theorem

The net work done on an object equals the change in kinetic energy:

$$W_{\text{net}} = \Delta E_k = E_{k,f} - E_{k,i}.$$

Example 7.3: Work needed to speed up a car

A 1000 kg car speeds up from 10 m/s to 20 m/s. Find the net work done.

$$E_{k,i} = \frac{1}{2} \cdot 1000 \cdot 10^2 = 50000\,\mathrm{J},$$

$$E_{k,f} = \frac{1}{2} \cdot 1000 \cdot 20^2 = 200000\,\mathrm{J}.$$

$$W_{\text{net}} = 200000 - 50000 = 150000\,\mathrm{J}.$$

7.4 Potential energy

Gravitational potential energy near Earth's surface:

$$E_p = mgh.$$

Elastic potential energy of a spring:

$$E_s = \frac{1}{2}kx^2.$$

Mechanical energy

Mechanical energy is the sum of kinetic and potential energy:

$$E_{\text{mech}} = E_k + E_p + E_s + \cdots.$$

If only conservative forces act, mechanical energy is conserved:

$$E_{\text{initial}} = E_{\text{final}}.$$

Example 7.4: Falling object by energy

A stone of mass 1 kg falls from height 5 m. Ignore air resistance and use $g = 10\,\mathrm{m/s^2}$. Find speed just before hitting the ground.
Initial energy:

$$E_i = mgh = 1 \cdot 10 \cdot 5 = 50\,\mathrm{J}.$$

Final energy is kinetic:

$$E_f = \frac{1}{2}mv^2.$$

Energy conservation:

$$50 = \frac{1}{2} \cdot 1 \cdot v^2.$$

$$v^2 = 100, \qquad v = 10\,\mathrm{m/s}.$$

7.5 Power

Power

Power is work or energy transferred per unit time:

$$P = \frac{W}{t} = \frac{\Delta E}{t}.$$

Power is measured in watts:

$$1\,\mathrm{W} = 1\,\mathrm{J/s}.$$

Example 7.5: Power of lifting

A machine lifts a 20 kg object by 3 m in 4 s. Use $g = 10\,\mathrm{m/s^2}$.

$$\Delta E = mgh = 20 \cdot 10 \cdot 3 = 600\,\mathrm{J}.$$

$$P = \frac{600}{4} = 150\,\mathrm{W}.$$

Common mistake: energy is not a force

Energy and force have different units and meanings. Force is measured in newtons. Energy is measured in joules. Do not write equations like $E = ma$.

8 Momentum and collisions

Momentum is useful when forces during an interaction are large and short-lasting, such as during a collision.

Momentum

Momentum is mass times velocity:

$$\boldsymbol{p} = m\boldsymbol{v}.$$

It is a vector.

8.1 Impulse

Impulse

Impulse is force times time:

$$\boldsymbol{J} = \boldsymbol{F}\Delta t.$$

Impulse equals change in momentum:

$$\boldsymbol{J} = \Delta\boldsymbol{p}.$$

Example 8.1: Impulse

A force of 30 N acts on a ball for 0.2 s. Find impulse.

$$J = F\Delta t = 30 \cdot 0.2 = 6\,\text{N.s}.$$

Since 1 N.s = 1 kg.m/s, the momentum changes by 6 kg.m/s.

8.2 Conservation of momentum

Conservation of momentum

If the external net force on a system is zero, total momentum is conserved:

$$\boldsymbol{p}_{\text{initial}} = \boldsymbol{p}_{\text{final}}.$$

For two objects in one dimension:

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}.$$

8.3 Perfectly inelastic collision

In a perfectly inelastic collision, objects stick together.

Example 8.2: Objects stick together

A 2 kg cart moving at 6 m/s collides with a 4 kg cart at rest. They stick together. Find final speed.
Initial momentum:

$$p_i = 2 \cdot 6 + 4 \cdot 0 = 12\,\text{kg.m/s}.$$

Final mass:

$$m_{\text{total}} = 2 + 4 = 6\,\text{kg}.$$

Momentum conservation:

$$12 = 6v_f.$$

$$v_f = 2\,\text{m/s}.$$

8.4 Elastic collision

In an elastic collision, momentum and kinetic energy are both conserved. In high-school problems, elastic collisions are often one-dimensional.

Collision types

- **Elastic**: momentum conserved, kinetic energy conserved.
- **Inelastic**: momentum conserved, kinetic energy not conserved.
- **Perfectly inelastic**: objects stick together.

Common mistake: conserving kinetic energy in every collision

Momentum is conserved in isolated collisions. Kinetic energy is conserved only in elastic collisions.

9 Circular motion

Circular motion is motion along a circular path. Even if speed is constant, velocity changes because direction changes.

9.1 Period, frequency, and speed

Period and frequency

The period T is the time for one full revolution. Frequency f is the number of revolutions per second:

$$f = \frac{1}{T}.$$

For circular motion with radius r, speed is

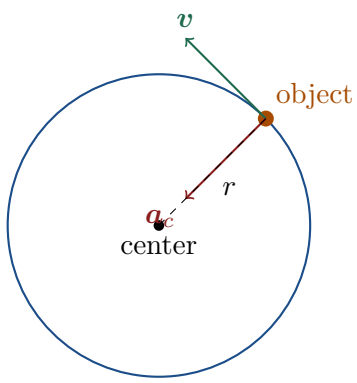
$$v = \frac{2\pi r}{T} = 2\pi r f.$$

9.2 Centripetal acceleration

Centripetal acceleration

In circular motion, acceleration toward the center is

$$a_c = \frac{v^2}{r}.$$



9.3 Centripetal force

Centripetal force is not a new kind of force. It is the name for the net force directed toward the center:

$$F_c = ma_c = m\frac{v^2}{r}.$$

Depending on the situation, the centripetal force may be caused by tension, gravity, friction, or a normal force.

Example 9.1: Stone on a string

A 0.5 kg stone moves in a horizontal circle of radius 1 m with speed 4 m/s. Find the centripetal force.

$$F_c = m\frac{v^2}{r} = 0.5 \cdot \frac{4^2}{1} = 8\,\mathrm{N}.$$

The string must provide 8 N of inward force.

Example 9.2: Car turning on a flat road

A car of mass 1000 kg takes a turn of radius 50 m at speed 10 m/s. Find the required friction force.

$$F_c = m\frac{v^2}{r} = 1000 \cdot \frac{10^2}{50} = 2000\,\mathrm{N}.$$

Static friction between tires and road provides this inward force.

Common mistake: drawing a force outward

For an object actually moving in a circle, the net force points inward. The feeling of being pushed outward in a turning car is due to inertia, not an outward physical force in an inertial frame.

10 Gravitation and orbits

Gravity is the attractive interaction between masses.

10.1 Newton's law of universal gravitation

Universal gravitation

Two masses attract each other with force

$$F = G\frac{m_1 m_2}{r^2},$$

where G is the gravitational constant and r is the distance between their centers.

Near Earth's surface, this complicated-looking law becomes the familiar weight formula:

$$F_g = mg.$$

10.2 Weight and mass

Mass measures how much matter an object has and how hard it is to accelerate it. Weight is the gravitational force on the object.

Quantity	Meaning	Unit
Mass m	amount of matter/inertia	kg
Weight mg	gravitational force	N

10.3 Orbital motion

A satellite in a circular orbit is constantly falling toward Earth, but also moving sideways fast enough to keep missing Earth.

For a circular orbit, gravity provides centripetal force:

$$G\frac{Mm}{r^2} = m\frac{v^2}{r}.$$

The satellite mass m cancels:

$$v = \sqrt{\frac{GM}{r}}.$$

Example 10.1: Why satellites do not need engines all the time

In an ideal circular orbit without air resistance, gravity is the centripetal force. It changes the direction of velocity but does not reduce the speed. Therefore a satellite can continue orbiting without an engine constantly pushing it forward.

10.4 Kepler's laws

Kepler's laws describe planetary motion:

1. Planets move in ellipses with the Sun at one focus.
2. A line from the Sun to a planet sweeps equal areas in equal times.
3. The square of orbital period is proportional to the cube of orbital size:

$$T^2 \propto r^3.$$

For high-school mechanics, the main lesson is that gravity explains both falling apples and planetary motion.

11 Oscillations

Oscillation is repeated motion around an equilibrium position. Examples: a mass on a spring, a pendulum, a vibrating string.

11.1 Spring force

Hooke's law

For a spring stretched or compressed by displacement x, the spring force has magnitude

$$F_s = kx.$$

The force points toward equilibrium.

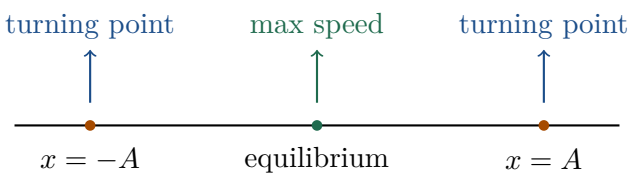
The constant k is the spring constant. A larger k means a stiffer spring.

11.2 Energy in a spring

Elastic potential energy:

$$E_s = \frac{1}{2}kx^2.$$

At maximum displacement, speed is zero and energy is stored in the spring. At equilibrium, spring potential energy is zero and kinetic energy is maximum.



11.3 Period of a mass-spring oscillator

For an ideal mass-spring system:

$$T = 2\pi\sqrt{\frac{m}{k}}.$$

For a small-angle pendulum:

$$T = 2\pi\sqrt{\frac{\ell}{g}}.$$

Example 11.1: Spring energy

A spring with $k = 200\,\mathrm{N/m}$ is compressed by $0.10\,\mathrm{m}$. Find stored energy.

$$E_s = \frac{1}{2}kx^2 = \frac{1}{2} \cdot 200 \cdot (0.10)^2 = 1\,\mathrm{J}.$$

Example 11.2: Pendulum period

A pendulum has length $\ell = 1\,\mathrm{m}$. Use $g \approx 9.81\,\mathrm{m/s^2}$.

$$T = 2\pi\sqrt{\frac{1}{9.81}} \approx 2.0\,\mathrm{s}.$$

12 Fluids: pressure and buoyancy

Fluids include liquids and gases. They can flow and exert pressure.

12.1 Pressure

Pressure

Pressure is force per unit area:

$$p = \frac{F}{A}.$$

The SI unit is the pascal:

$$1\,\mathrm{Pa} = 1\,\mathrm{N/m^2}.$$

A smaller area gives larger pressure for the same force. This is why a sharp knife cuts better than a dull one.

12.2 Hydrostatic pressure

In a liquid at rest, pressure increases with depth:

$$p = p_0 + \rho g h.$$

Here:

- p_0 is pressure at the surface,
- ρ is density of the liquid,
- h is depth.

Example 12.1: Pressure under water

Find gauge pressure at depth $h = 2\,\mathrm{m}$ in water. Use $\rho = 1000\,\mathrm{kg/m^3}$ and $g = 10\,\mathrm{m/s^2}$.

$$\rho g h = 1000 \cdot 10 \cdot 2 = 20000\,\mathrm{Pa}.$$

This is pressure above atmospheric pressure.

12.3 Buoyancy

Archimedes' principle

An object in a fluid experiences an upward buoyant force equal to the weight of the displaced fluid:

$$F_b = \rho_{\text{fluid}} g V_{\text{displaced}}.$$

An object floats if buoyant force can balance its weight.

Example 12.2: Buoyant force

A submerged object displaces $0.003\,\mathrm{m^3}$ of water. Find buoyant force. Use $\rho = 1000\,\mathrm{kg/m^3}$ and $g = 10\,\mathrm{m/s^2}$.

$$F_b = \rho g V = 1000 \cdot 10 \cdot 0.003 = 30\,\mathrm{N}.$$

13 A compact formula sheet

Kinematics

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t}$$
$$a_{\text{avg}} = \frac{\Delta v}{\Delta t}$$
$$v = v_0 + at$$
$$x = x_0 + v_0 t + \frac{1}{2}at^2$$
$$v^2 = v_0^2 + 2a(x - x_0)$$

Projectile motion

$$v_{0x} = v_0 \cos\theta$$
$$v_{0y} = v_0 \sin\theta$$
$$x = v_{0x} t$$
$$y = v_{0y} t - \frac{1}{2}gt^2$$
$$R = \frac{v_0^2 \sin 2\theta}{g}$$

Forces

$$\sum F = ma$$
$$F_g = mg$$
$$f_k = \mu_k N$$
$$F_s = kx$$

Energy

$$W = Fs\cos\theta$$
$$E_k = \frac{1}{2}mv^2$$
$$E_p = mgh$$
$$E_s = \frac{1}{2}kx^2$$
$$P = \frac{W}{t}$$

Momentum

$$p = mv$$

$$J = F\Delta t = \Delta p$$

$$p_i = p_f$$

Circular motion

$$v = \frac{2\pi r}{T}$$

$$a_c = \frac{v^2}{r}$$

$$F_c = m\frac{v^2}{r}$$

Gravitation

$$F = G\frac{m_1 m_2}{r^2}$$

$$v_{\text{orbit}} = \sqrt{\frac{GM}{r}}$$

Fluids

$$p = \frac{F}{A}$$

$$p = p_0 + \rho g h$$

$$F_b = \rho g V$$

14 Worked mixed problems

Problem 1: A two-step kinematics problem

A car starts from rest, accelerates at $3\,\mathrm{m/s^2}$ for $5\,\mathrm{s}$, then continues at constant speed for $10\,\mathrm{s}$. Find total distance.

Step 1: Acceleration phase.

$$v = 0 + 3 \cdot 5 = 15\,\mathrm{m/s}.$$

$$s_1 = \frac{1}{2} \cdot 3 \cdot 5^2 = 37.5\,\mathrm{m}.$$

Step 2: Constant speed phase.

$$s_2 = vt = 15 \cdot 10 = 150\,\mathrm{m}.$$

Total distance.

$$s = s_1 + s_2 = 37.5 + 150 = 187.5\,\mathrm{m}.$$

Problem 2: Dynamics with friction

A $5\,\mathrm{kg}$ block is pushed with horizontal force $25\,\mathrm{N}$ on a floor. The coefficient of kinetic friction is 0.3. Use $g = 10\,\mathrm{m/s^2}$. Find acceleration.

$$N = mg = 5 \cdot 10 = 50\,\mathrm{N}.$$

$$f_k = \mu_k N = 0.3 \cdot 50 = 15\,\mathrm{N}.$$

$$F_{\text{net}} = 25 - 15 = 10\,\mathrm{N}.$$

$$a = \frac{F_{\text{net}}}{m} = \frac{10}{5} = 2\,\mathrm{m/s^2}.$$

Problem 3: Energy on a slide

A child starts from rest at the top of a frictionless slide of height $3\,\mathrm{m}$. Find speed at the bottom. Use $g = 10\,\mathrm{m/s^2}$.

At the top:

$$E_i = mgh.$$

At the bottom:

$$E_f = \frac{1}{2}mv^2.$$

Conservation:

$$mgh = \frac{1}{2}mv^2.$$

Mass cancels:

$$gh = \frac{1}{2}v^2.$$

$$v = \sqrt{2gh} = \sqrt{2 \cdot 10 \cdot 3} = \sqrt{60} \approx 7.75\,\mathrm{m/s}.$$

Problem 4: Collision

A 1 kg ball moving right at 8 m/s sticks to a 3 kg cart at rest. Find final speed.
Momentum conservation:

$$1 \cdot 8 + 3 \cdot 0 = (1+3)v_f.$$

$$8 = 4v_f, \qquad v_f = 2\,\mathrm{m/s}.$$

The combined object moves right at 2 m/s.

Problem 5: Circular motion

A toy car moves in a circle of radius 0.5 m at speed 2 m/s. Its mass is 0.2 kg. Find centripetal acceleration and force.

$$a_c = \frac{v^2}{r} = \frac{2^2}{0.5} = 8\,\mathrm{m/s^2}.$$

$$F_c = ma_c = 0.2 \cdot 8 = 1.6\,\mathrm{N}.$$

15 Practice set

A. Kinematics

1. A train moves at constant speed 25 m/s for 2 min. How far does it travel?
2. A ball accelerates from 4 m/s to 16 m/s in 6 s. Find acceleration.
3. A car brakes from 30 m/s to rest with acceleration $-6\,\mathrm{m/s^2}$. Find braking time and distance.
4. A stone is dropped from a height of 45 m. Use $g = 10\,\mathrm{m/s^2}$. Find fall time.

B. Forces

1. Find the weight of a 70 kg person. Use $g = 10\,\mathrm{m/s^2}$.
2. A 4 kg object has net force 12 N. Find acceleration.
3. A 20 kg box is pulled with 100 N on a frictionless floor. Find acceleration.
4. The same box has kinetic friction 30 N. Find acceleration.

C. Energy

1. Find kinetic energy of a 3 kg object moving at 4 m/s.
2. Find gravitational potential energy of a 2 kg object at height 5 m using $g = 10\,\mathrm{m/s^2}$.
3. A spring with $k = 100\,\mathrm{N/m}$ is stretched by 0.20 m. Find elastic energy.
4. A motor does 900 J of work in 3 s. Find power.

D. Momentum and circular motion

1. A 0.5 kg ball moves at 12 m/s. Find momentum.
2. A 2 kg cart moving at 5 m/s sticks to a 3 kg cart at rest. Find final speed.
3. An object moves in a circle of radius 2 m at speed 6 m/s. Find centripetal acceleration.
4. A 1.5 kg object has centripetal acceleration $8\,\mathrm{m/s^2}$. Find centripetal force.

16 Answers to selected exercises

Section 2 exercises

1. 5 m/s
2. 2 m/s^2
3. 15 m/s, 75 m
4. 2 s

Practice A

1. 3000 m
2. 2 m/s^2
3. 5 s, 75 m
4. 3 s

Practice B

1. 700 N
2. 3 m/s^2
3. 5 m/s^2
4. 3.5 m/s^2

Practice C

1. 24 J
2. 100 J
3. 2 J
4. 300 W

Practice D

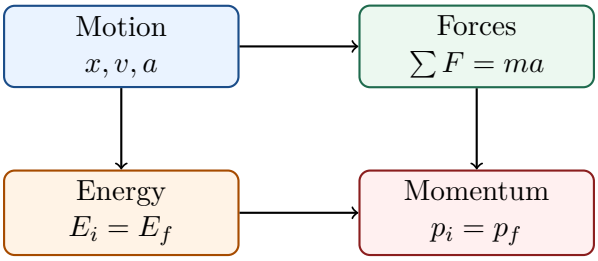
1. 6 kg.m/s
2. 2 m/s
3. 18 m/s^2
4. 12 N

17 Final summary

Mechanics is built from a small number of powerful ideas.

The big map of mechanics

- **Kinematics**: describes motion using x, v, and a.
- **Newton's laws**: connect forces to acceleration using $\sum F = ma$.
- **Energy**: compares beginning and ending states using E_k, E_p, work, and power.
- **Momentum**: handles collisions and short interactions using $p = mv$.
- **Circular motion**: velocity changes direction, so inward acceleration is required.
- **Gravity**: explains falling objects and orbits using the same physical idea.

The main challenge in mechanics is not memorizing formulas. The main challenge is choosing the right model. Before solving a problem, ask:

1. Is acceleration constant? Use kinematics.
2. Are forces important? Use Newton's laws.
3. Are initial and final states enough? Use energy.
4. Is there a collision or explosion? Use momentum.
5. Is motion circular? Look for inward net force.